

PENHAS DOURADAS, Portugal

WMO: 085680

Lat: 40.4114N

Lon: 7.5586W

Elev: 1381

StdP: 85.80

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.0	(c) -3.5	(d) -20.0	(e) 0.8	(f) 6.9	(g) -16.0	(h) 1.1	(i) 6.8	(j) 18.6	(k) 2.2	(l) 17.1	(m) 2.1	(n) 4.9	(o) 130	(p) 0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.9	(c) 27.5	(d) 15.4	(e) 25.9	(f) 14.8	(g) 24.5	(h) 14.5	(i) 17.1	(j) 23.9	(k) 16.4	(l) 23.2	(m) 15.6	(n) 22.4	(o) 3.6	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.7	12.4	19.7	13.8	11.7	18.8	13.1	11.1	18.1	53.7	24.0	51.2	23.2	48.8	22.4	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	13.6	11.7	-7.1	30.1	2.0	1.2	-8.5	31.0	-9.7	31.7	-10.9	32.4	-12.4	33.3
			-8.1	18.8	2.1	1.1	-9.6	19.6	-10.9	20.3	-12.0	20.9	-13.6	21.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	3.4	3.8	5.6	6.9	10.4	14.9	17.8	18.4	15.3	10.8	5.7	4.4
	DBStd	6.65	3.49	3.66	3.99	4.30	4.61	4.58	4.22	3.94	3.71	3.88	3.35	3.18
	HDD10.0	1055	206	174	144	114	53	11	1	3	36	135	176	
	HDD18.3	3236	464	406	393	343	248	119	62	49	103	234	381	433
	CDD10.0	988	1	1	9	22	66	158	243	259	162	61	4	2
	CDD18.3	128	0	0	0	0	2	17	46	50	12	1	0	0
	CDH23.3	571	0	0	0	0	5	56	219	256	34	2	0	0
	CDH26.7	81	0	0	0	0	0	3	31	44	3	0	0	0
(14) Wind	WSAvg	4.8	5.0	5.1	5.3	5.3	4.5	4.6	5.0	4.4	3.9	4.6	5.2	5.2
(15) Precipitation	PrecAvg	1670	225	201	134	143	123	71	22	17	69	184	241	247
	PrecMax	2669	541	541	516	441	237	255	97	77	249	471	686	803
	PrecMin	1024	0	20	0	22	12	5	0	0	2	5	4	31
	PrecStd	421	146	152	116	92	71	68	25	19	54	126	176	196
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	14.0	17.5	19.7	24.0	27.0	29.1	30.3	26.2	23.0	15.6	14.4
		MCWB	5.7	6.1	8.5	11.4	14.0	15.3	15.9	15.9	14.3	12.0	9.1	6.4
	2%	DB	11.2	12.2	15.1	17.7	21.4	24.9	27.5	27.8	24.1	20.1	13.3	12.0
		MCWB	4.7	6.0	7.5	10.5	12.8	14.7	15.6	15.4	13.8	11.2	7.9	5.4
	5%	DB	9.4	10.5	13.2	15.7	19.5	23.5	25.9	26.1	22.6	17.9	11.5	10.4
		MCWB	4.5	5.0	6.8	9.5	12.1	14.3	15.2	14.8	13.1	10.6	7.5	5.1
	10%	DB	8.0	8.8	11.4	13.7	17.6	21.7	24.2	24.5	21.0	16.1	10.2	8.8
		MCWB	4.6	4.4	5.9	8.4	11.2	13.8	14.6	14.5	12.7	10.1	7.2	5.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	9.0	10.1	12.9	15.9	17.6	18.4	17.8	16.3	14.7	12.0	9.3
		MCDB	10.1	10.9	14.8	18.3	21.1	23.2	25.5	25.6	22.1	18.0	13.4	10.6
	2%	WB	7.7	7.6	8.9	11.2	14.3	16.6	17.3	16.7	15.2	13.5	10.4	8.3
		MCDB	8.4	9.7	12.3	16.2	19.3	22.4	24.3	24.3	21.2	16.4	11.4	9.3
	5%	WB	6.7	6.6	7.9	10.0	13.0	15.6	16.4	16.1	14.4	12.5	9.3	7.4
		MCDB	7.6	8.6	11.2	14.8	18.2	21.5	23.3	23.6	20.2	15.3	10.2	8.5
	10%	WB	5.8	5.6	7.0	8.9	11.7	14.6	15.6	15.3	13.6	11.6	8.2	6.4
		MCDB	6.9	7.5	10.1	13.1	16.6	20.5	22.5	22.8	19.0	14.3	9.3	7.6
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.0	6.6	7.0	7.4	8.0	8.8	8.9	7.8	5.9	5.1	5.3
		MCDBR	7.1	7.7	8.7	8.9	9.1	9.1	9.5	9.7	9.1	7.9	6.7	6.9
		MCWBR	5.4	5.7	6.1	5.9	6.2	6.7	6.3	5.6	5.4	5.6	5.6	5.7
	5% WB	MCDBR	5.5	6.5	7.6	8.5	8.6	8.8	8.9	9.1	8.3	6.3	5.2	5.6
		MCWBR	5.1	5.6	6.0	6.1	6.2	7.0	6.7	6.2	5.6	5.1	5.0	5.4
(40) Clear-Sky Solar Irradiance	taub	0.265	0.276	0.301	0.304	0.310	0.314	0.304	0.315	0.306	0.297	0.275	0.264	
	taud	2.602	2.574	2.515	2.510	2.512	2.515	2.522	2.480	2.537	2.585	2.603	2.602	
	Ebn at Noon	922	956	959	972	965	958	964	947	940	917	901	896	
	Edh at Noon	67	80	95	102	104	104	103	104	92	78	67	62	
(44) All-Sky Solar Radiation	RadAvg	2.01	3.00	4.05	5.13	6.36	7.29	7.69	6.81	5.18	3.33	2.20	1.80	
	RadStd	0.31	0.43	0.60	0.65	0.69	0.43	0.25	0.31	0.40	0.39	0.39	0.27	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+51

Nomenclature: See separate page